Lecture plan · for the lecturer

Designing an IoT capstone

IoT · University · Lecture 4

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Scaffold a lab sequence.
- Run labs safely and cheaply.
- Assess teams fairly.

Activity: Map skills to labs

1. Warm-up (5 min): Show the week-six progress chart and ask colleagues what they would change.
2. List (10 min): Colleagues list every skill their capstone needs.
3. Map (20 min): Map each skill to an earlier lab and mark gaps.
4. Draft (20 min): Write five criteria and four levels, including peer assessment.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which skill had no lab?
- What is your biggest safety risk?
- How will you spot a passenger in a team?

Exit ticket

- Name one lab you will add and the capstone skill it builds.

Try it in your next lecture
List the skills your capstone needs, check that an earlier lab teaches each one, and add a lab for any that is missing.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **One of four teammates did almost nothing. Best evidence for adjusting marks?**

◯ Structured peer assessment plus commit and test logs ◯ The lecturer's impression

◯ The final demo

2 **Plan the weeks.**

A 14-week module with 8 lab weeks and 1 proposal week leaves _____ weeks for the capstone. With a demo at the midpoint of the capstone, it falls about _____ of its weeks in.

Word bank: 6 · 5 · 2 · 4

3 **Which evidence should a capstone rubric assess beyond the demo? Choose all that apply.**

☐ A test log against requirements ☐ Version-control history ☐ A measured power budget

☐ The price of the parts

4 **Giving the open capstone in week one maximises the time students spend learning.**

◯ True ◯ False

5 **Put the lab sequence from the video in order.**

☐ Low-power modes

☐ Networking

☐ Sensors and the ADC

☐ Digital input and output

☐ Timers and interrupts

Answers · for the lecturer

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1 **Structured peer assessment plus commit and test logs**

Several independent records of contribution are fairer than one impression.

2 **A 14-week module with 8 lab weeks and 1 proposal week leaves 5 weeks for the capstone. With a demo at the midpoint of the capstone, it falls about 2 of its weeks in.**

14 − 8 − 1 = 5, and about halfway through 5 weeks is week 2 to 3.

3 **A measured power budget; A test log against requirements; Version-control history**

These show engineering process and individual contribution; part price says little about learning.

4 **False**

Without prior skills, early time goes on configuration and debugging; scaffolded labs first make capstone time productive.

5 **Top to bottom: 4, 5, 2, 1, 3**

This is the sequence from the video and the reading: each lab produces a part the capstone reuses.

✎ **For you to think about**

Which skill do your capstone teams most often lack? Which earlier lab would build it?